

Beyond Accuracy: A Multi-Axis Evaluation Framework for Interpretable Topic Models on Hyperspectral Imagery

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CAOS open-research programme · CAOS_LDA_HSI, interpretable topic models for hyperspectral imagery

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_LDA_HSI

Abstract. Topic models, in particular Latent Dirichlet Allocation (LDA), have been adapted to hyperspectral imagery (HSI) as a route to *interpretable* spectral mixtures: each spectral unit (canonically a pixel) becomes a document of quantised band-frequency tokens and the inferred topic–word distributions act as non-negative basis spectra. Yet the dominant evaluation criterion remains downstream classification accuracy, an accuracy-only view that is silent on whether the basis is reproducible, band-robust, calibrated, or actually outperforms standard linear baselines. We propose and instantiate a twelve-axis evaluation framework for LDA-style topic models on HSI that quantifies, on the same set of benchmarks: (F-1) classification macro-F1 compared across five method families (paired differences and a hierarchical Bayesian model), (F-2) topic–word coherence (c_v , c_{NPMI} , U-Mass), (F-3) seed stability for both LDA and two neural variants (ProdLDA, ETM), (F-4) capacity sensitivity across $K \in \{4, 6, 8, 10, 12, 16\}$, (F-5) band-mask robustness, (F-6) cross-method clustering agreement (LDA vs Felzenszwalb, SLIC, patches, pixels), (F-7) topic–label coupling via $P(L | t)$ entropy and KL, (F-8) per-topic Hungarian-aligned identity tracking, (F-9) HIDSAG cross-preprocessing stability, (F-10) cross-scene topic transfer, (F-11) rate–distortion of the topic representation, and (F-12) external alignment of the topics with a spectral library and a method–rank cross-comparison. We apply the framework to six standard HSI benchmarks (Indian Pines, Salinas, Salinas-A, Pavia University, Kennedy Space Center, Botswana) and five subsets of the HIDSAG mineralogical database (GEOMET, MINERAL1, MINERAL2, GEOCHEM, PORPHYRY). The framework yields three findings that the accuracy-only view does not surface. *First*, on the labelled scenes the topic-routed soft classifier, which routes the raw spectrum to per-topic logistic experts, matches the raw-spectrum logistic baseline in macro-F1 (0.915 against 0.910, a paired difference of 0.005), while a logistic regression on the topic proportions θ alone trails the baseline by 0.325: the topics add an interpretable routing to the raw spectrum but do not replace it. *Second*, on Salinas-A under the SWIR-only band mask the paired ARI between canonical and masked dominant-topic maps reaches 0.766, while on Kennedy Space Center under every mask, and on Botswana under the SWIR mask, the same paired ARI collapses to ≈ 0.01 , separating scenes on which an interpretability claim is band-robust from scenes on

which it is not. *Third*, the θ -only deficit holds on HIDSAG: over its nineteen mineralogical targets the topic-logistic classifier trails raw-logistic by 0.268 macro-F1 (lower on 18 of 19 targets), contrary to the prima-facie expectation that mineralogical mixtures favour topic decomposition over the raw spectrum. We release 3895 deterministic derived artefacts (449 MB), the builder source, the FastAPI backend, the React frontend, and a 109-endpoint test harness as a public, MIT-licensed reproducibility package (https://github.com/fsantibanezleal/CAOS_LDA_HSI; live at <https://lda-hsi.fasl-work.com>).

Index Terms. hyperspectral imagery, latent Dirichlet allocation, topic models, interpretability, reproducibility, evaluation framework, neural topic models, ProdLDA, ETM, hierarchical Bayesian comparison, band-mask robustness, Hungarian topic alignment, adjusted Rand index, mineralogical hyperspectral data.

I. INTRODUCTION

HYPERSPECTRAL imagery (HSI) records pixel-wise reflectance across hundreds of contiguous narrow bands, producing a spectral signature for each spatial location that, in principle, admits material-level discrimination from a single sensor pass [1], [2]. The dominant analysis paradigms on HSI today are *linear unmixing*, which decomposes a pixel into a small set of endmembers and their abundances [3], [4], and *deep spectral–spatial classification*, in which 1-D or 3-D convolutional networks map a pixel and its neighbourhood directly to a class label [5], [6]. Both paradigms have made substantial progress on the canonical labelled benchmarks (Indian Pines, Salinas, Pavia University), and both have well-understood inductive biases (linear additivity for unmixing; high-capacity non-linear feature extraction for deep models).

A third, less developed paradigm treats each pixel as a *document* of quantised band-frequency tokens and applies a topic model, typically Latent Dirichlet Allocation (LDA) [7], [8], to recover a basis of topic–word distributions ϕ_k and a per-pixel posterior simplex θ_d . The promise is interpretability: ϕ_k is a non-negative spectral basis that can be inspected as a basis spectrum and visually or numerically compared with reference libraries; θ_d is a soft mixture over K basis spectra that can be inspected as an abundance vector or fed into a downstream classifier as a low-dimensional feature [9]. The

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All code, derived artefacts, and the interactive web application are released under MIT at https://github.com/fsantibanezleal/CAOS_LDA_HSI.

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paradigm has been extended to neural-topic-model variants such as ProdLDA [10] and the Embedded Topic Model (ETM) [11], which replace the variational EM of online LDA with an amortised variational autoencoder [12], [13] and a learned embedding basis, respectively.

A. The accuracy-only view and its limits

The empirical literature evaluating LDA-style topic models on HSI is small but converging on a familiar pattern: a paper proposes a topic model variant, applies it to one or two labelled scenes, and reports overall accuracy (OA) and kappa from a downstream classifier built on θ against published OA/kappa numbers for a chosen deep or classical baseline. We refer to this as the *accuracy-only view*. The accuracy-only view is silent on questions that are central to the interpretability claim that motivates using topic models in the first place:

- Is the topic basis ϕ reproducible under different random seeds? If not, the “interpretable mineral signature on topic 3” narrative is a description of the realisation, not of the model.
- Is ϕ robust to the choice of *spectral window*? A topic basis that flips identity when one removes the atmospheric water bands provides weak grounds for any spectral-window-conditional interpretation.
- Is the posterior θ *well calibrated*, in the sense that pixels with high entropy in θ correspond to materially mixed regions and pixels with low entropy correspond to materially homogeneous regions?
- Does the topic representation *actually outperform* the raw-spectrum logistic baseline once one accounts for the multi-scene multi-fold variability via a proper hierarchical model rather than a single OA average?
- Does the dominant-topic map *spatially agree* with unsupervised segmentation methods (SLIC, Felzenszwalb) that operate on raw spectra without ever fitting a topic model, and if so does the agreement reflect signal or shared smoothness bias?

A single accuracy number cannot answer any of these. A paper that reports $OA = 0.85$ on Indian Pines might be reporting on a band-fragile, seed-unstable, mode-collapsed topic basis whose downstream classifier happens to predict the same labels as a linear baseline on the raw spectrum. The interpretability that initially motivated choosing a topic model has, in that scenario, evaporated.

B. Contributions

This paper proposes and instantiates a twelve-axis evaluation framework for LDA-style topic models on HSI. The framework is designed so that each axis is operational (deterministic, with a fixed-seed reproducibility contract), comparable across scenes, and surfaces a specific failure mode that the accuracy-only view hides. Concretely:

- 1) We define twelve evaluation axes (F-1 through F-12) covering classification, coherence, seed stability, capacity, band robustness, cross-method agreement, topic-label coupling, topic identity tracking, preprocessing ro-

bustness, cross-scene transfer, rate-distortion, and external spectral-library alignment. Each axis specifies (a) the metric, (b) the unit of comparison, (c) the deterministic builder used to compute it, and (d) the JSON artefact included in the public release.

- 2) We apply the framework to six standard labelled HSI benchmarks and five HIDSAG mineralogical subsets. All numerical results are computed from the same canonical LDA fit (K chosen by the rule $K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$, $T_{\text{max}} = 60$ online-VB passes, seed 42, samples per class 220).
- 3) We compare against two neural topic models (ProdLDA, ETM) on the labelled scenes with the same K and the same five-seed stability protocol, surfacing a head-to-head accuracy ordering (LDA > ETM > ProdLDA) that runs counter to the neural-supremacy expectation.
- 4) We compare against three non-topic baselines (raw-spectrum logistic regression, PCA- K logistic regression, theta-only logistic regression) and two topic-routed classifiers (hard- and soft-gated mixtures of per-topic logistic experts). Methods are compared by their paired macro-F1 differences over the (scene, fold) cells and by the pairwise $P[A > B]$ of a hierarchical Bayesian model that pools over scenes and folds, instead of a single-OA bar chart.
- 5) We show that the topic proportions alone discard class information on both benchmark families: a logistic regression on θ trails the raw-spectrum baseline by 0.325 macro-F1 on the labelled scenes and by 0.268 on the HIDSAG targets, while the topic-routed classifier, which keeps the raw spectrum, matches the baseline on the labelled scenes.
- 6) We release 3895 deterministic derived artefacts (449 MB), the builder source code (Python, NumPy, scikit-learn, Gensim, PyTorch), the FastAPI backend, the React frontend, and a 109-endpoint test harness as a public MIT-licensed package.

C. Paper organisation

Section II surveys related work on LDA for HSI, on neural topic models, and on accuracy-beyond-OA evaluation in remote sensing. Section III introduces the twelve-axis framework and the canonical fit. Section IV states the methodological details of each axis (F-1 through F-12). Section V reports the experimental setup (scenes, HIDSAG subsets, tokenisation, hyperparameters). Section VI reports the per-axis results. Section VII discusses interpretation and limitations. Section VIII documents the reproducibility artefact set. Section IX concludes.

II. RELATED WORK

A. LDA-style topic models on hyperspectral imagery

LDA [7] was originally proposed as a generative model for text corpora; its application to HSI requires tokenising the continuous reflectance spectrum into a discrete band-frequency vocabulary. Zou and Zare [9] applied *PM-LDA* (partial-membership LDA), in which each pixel admits a soft

membership to multiple topics representing spectral classes, to hyperspectral unmixing with endmember variability. An adjacent line by Wahabzada et al. [14] applied probabilistic topic models to hyperspectral plant phenotyping to recover an interpretable “hyperspectral language of plants”. Beyond these direct applications, the broader HSI literature treats θ implicitly through related abstractions: linear unmixing [3], [4], [15] estimates a per-pixel abundance over a small set of endmembers, which is functionally analogous to θ over a basis of ϕ_k ; the survey of Borsoi et al. [15] consolidates the modern spectral-variability variants and the Samson / Jasper-Ridge / Urban benchmarks that the unmixing community uses. Sparse coding and dictionary learning on spectra are similarly analogous. The conceptual gap addressed by this paper is *not* a new tokenisation or a new variant of LDA; it is the absence of a shared, axis-by-axis evaluation framework that lets follow-on work be compared on a uniform basis.

B. Neural topic models

ProdLDA [10] replaces the variational-EM inference of LDA with a Laplace-approximated VAE [12], yielding amortised inference and substantially sharper topic posteriors on text. The Embedded Topic Model (ETM) [11] additionally embeds words and topics into a shared continuous space, allowing rare words to inherit semantic neighbours’ topic associations. The model run in this paper under the ProdLDA label is the mixture-decoder form of Srivastava and Sutton’s amortised topic model (their NVLDA): a standard normal prior $\delta_d \sim \mathcal{N}(0, I)$, $\theta_d = \text{softmax}(\delta_d)$ and $w_{dn} \sim \text{Cat}(\sum_k \theta_{dk} \beta_k)$ with $\beta_k = \text{softmax}(\omega_k)$ from a free $K \times |V|$ matrix of logits ω_k ; the product-of-experts decoder that defines ProdLDA was not run. ETM, as run here, has the same prior and amortised encoder and the decoder $\beta_k = \text{softmax}(\rho^\top \alpha_k)$ with word embeddings of dimension $L = 64$ and a mixture likelihood as in [11], so the two neural models differ only in the decoder parameterisation (a free topic–word matrix against a rank- L factorisation). Miao et al. [13] and Higgins et al. [16] proposed related VAE-based topic and disentanglement variants. Card et al. [17] extended the family with SCHOLAR, a metadata-conditioned neural topic model that recovers the Dirichlet-multinomial-regression idea of Mimno and McCallum [18] in a neural form; we treat SCHOLAR as the natural deep-learning counterpart for future work on the HIDSAG continuous targets. Direct applications of these neural topic models to HSI are rare in the published literature; the small body of work that does compare them on HSI typically does so on a single scene with a single seed, leaving the seed-stability question open.

C. Beyond-OA evaluation in remote sensing

The remote-sensing community has independently recognised that overall accuracy alone is an inadequate measure of model quality. Röder et al. [19] surveyed coherence measures for topic models in NLP and argued that c_v and c_{NPMI} track human judgements of interpretability more reliably than perplexity. Hubert and Arabie [20] formalised the adjusted Rand index for clustering comparisons; Vinh et al. [21]

extended it with information-theoretic complements. Kuhn’s Hungarian algorithm [22] is the standard tool for solving the label-permutation problem when comparing two clusterings of the same data. Pineau et al. [23] argued that reproducibility checklists are a precondition for trustworthy beyond-accuracy reporting. Felzenszwalb and Huttenlocher [24] and Achanta et al. [25] provide spatial-segmentation baselines that serve as cross-method-agreement anchors in the framework below.

III. TWELVE-AXIS EVALUATION FRAMEWORK

A. Canonical fit

All twelve axes are computed against a single, deterministic *canonical fit* that we describe once and refer back to from each axis. The canonical fit operates as follows.

Data slice. For each labelled scene, we sample 220 pixels per class without replacement from the labelled-pixel mask, using NumPy’s `default_rng(42)`. Pavia U (9 classes) yields $D = 1980$ documents; Indian Pines (16 classes) yields $D = 2812$, sampling $\min(220, \text{class size})$ per class and retaining all 16 classes; Salinas yields $D = 3520$; Salinas-A (6 classes) yields $D = 1320$; KSC (13 classes) yields $D = 2686$; Botswana (14 classes) yields $D = 2775$. For each HIDSAG subset, the corpus is constructed from the public `hidsag_region_documents.npz` bundle, yielding $19\text{-target} \times 5\text{-subset}$ corpora over the `swir_low` modality.

Tokenisation. Each pixel spectrum is min–max normalised across bands and quantised into $Q + 1 = 13$ levels, producing integer counts $c_{d,b} \in \{0, \dots, 12\}$. The high-rate uniform-quantisation distortion bound $D \leq \frac{1}{12Q^2}$ [26] therefore yields an rms wordification noise below 0.025 in normalised reflectance units. The resulting document-term matrix has shape $D \times B$ where B is the band count (200 for Indian Pines, 204 for Salinas/Salinas-A, 103 for Pavia U, 176 for KSC, 145 for Botswana).

Model. Online variational Bayes LDA [8], scikit-learn implementation [27], with $K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$, $T_{\text{max}} = 60$, random seed 42, learning offset 10.0, learning decay 0.7. The resulting basis $\phi \in \mathbb{R}_+^{K \times B}$ and posterior $\theta \in \mathbb{R}_+^{D \times K}$ are serialised as a deterministic JSON blob plus a binary `theta_grid.bin` (float32 $H \times W \times K$) per scene.

B. The twelve axes

Each axis returns a single per-scene scalar (or a small vector) that can be reported in a table; jointly the twelve scalars constitute a *model card* for the topic basis on that scene. Table I lists the axes and the metric returned by each.

IV. METHOD: PER-AXIS DETAILS

A. F-1: classification across method families

The classical evaluation reports overall accuracy (OA) or kappa averaged over a small number of train/test folds. We

TABLE I
TWELVE-AXIS EVALUATION FRAMEWORK. EACH AXIS RETURNS A PER-SCENE METRIC, STORED AS A DETERMINISTIC JSON ARTEFACT UNDER DATA/DERIVED/.

Axis	Name	Primary metric	Builder	Artefact path
F-1	Classification	macro-F1; paired difference; $P[A > B]$	build_bayesian_*	method_statistics_labelled/
F-2	Coherence	c_v , c_{NPMI} , U-Mass	build_coherence	cross_method_agreement/
F-3	Seed stability	mean $\pm$ std ARI over 5 seeds	build_neural_topic_seed_stability	neural_topic_seed_stability/
F-4	Capacity sensitivity	ARI(K) curve	build_lda_sweep	lda_sweep/
F-5	Band-mask robustness	paired ARI	build_band_mask_canonical_comparison	band_masks/canonical_comparison.json
F-6	Cross-method agreement	ARI matrix	build_cross_method_agreement	cross_method_agreement/
F-7	Topic-label coupling	$H(L t)$, KL	build_topic_to_data	topic_to_data/
F-8	Per-topic identity	Hungarian σ^*	build_band_mask_canonical_comparison	same as F-5
F-9	Preprocessing stability	ARI across recipes	build_hidsag_cross_preprocessing_stability	hidsag_cross_preprocessing_stability/
F-10	Cross-scene transfer	transfer ARI	build_cross_scene_transfer	cross_scene_transfer/
F-11	Rate-distortion	MSE(θ) vs K	build_rate_distortion	rate_distortion_curve/
F-12	External alignment	library cosine; method rank	build_external_validation	external_validation/

instead fit a hierarchical Bayesian model over scene, fold, and method, following the structure

$$\text{score}_{m,s,f} = \mu_m + \delta_s + \rho_f + \epsilon_{m,s,f}, \quad (1)$$

$$\mu_m \sim \mathcal{N}(0, 1), \quad (2)$$

$$\delta_s \sim \mathcal{N}(0, 0.5), \quad (3)$$

$$\rho_f \sim \mathcal{N}(0, 0.2), \quad (4)$$

$$\epsilon_{m,s,f} \sim \mathcal{N}(0, \sigma), \quad \sigma \sim \text{HalfNormal}(0.5), \quad (5)$$

where m ranges over methods, s over scenes, f over folds, and score is the fold-level macro-F1. Sampling is performed with NUTS (1000 draws, 1000 tune, 2 chains) on the labelled-scene block; the HIDSAG block uses a parallel model with subsets in place of scenes and targets in place of folds. The archived fits record no convergence diagnostic such as the rank-normalised $\hat{R}$ of Vehtari et al. [28]. The model fixes no reference scene, so μ_m is identified only up to the scene offsets δ_s : every μ_m shares a common shift (about -0.17 from the observed mean on the labelled block) and a common posterior spread (about 0.19), and only differences between methods are identified. We therefore report the observed mean macro-F1 of each method, its paired difference from the raw-spectrum baseline over the cells (scene and fold, or subset and target) with a 94% bootstrap interval ($B = 5000$ resamples of the paired differences), and the pairwise probability $P[\mu_A > \mu_B]$ derived from the joint posterior. The model treats the residuals of different methods in the same cell as independent, so its $P[\mu_A > \mu_B]$ does not use the pairing; the folds of a scene share training pixels, so the paired intervals are in turn narrower than a resampling of independent cells would give. Five methods populate the labelled block: `pca_K_logistic` (PCA to K components followed by multinomial logistic regression), `raw_logistic` (multinomial logistic regression on the raw spectrum), `theta_logistic` (logistic on the topic poste-

rior θ), `topic_routed_hard` (hard-gated mixture of per-topic logistic experts), and `topic_routed_soft` (soft-gated mixture). The HIDSAG block uses an analogous five-method slate.

B. F-2: topic-word coherence

We compute three coherence measures per scene: the sliding-window c_v measure of Röder et al. [19], c_{NPMI} (normalised pointwise mutual information, following Newman et al. [29] and the NPMI formulation surveyed in [19]), and the intrinsic U-Mass measure of Mimno et al. [30]. The reference corpus for the co-occurrence estimate is the canonical document-term matrix of the same scene. Coherence is reported with top- $N = 15$ words per topic, averaged over K topics.

C. F-3: seed stability

For each labelled scene, we fit ProdLDA and ETM (in the form stated in Section II) with five random seeds, `seed` $\in \{0, 1, 2, 3, 4\}$, at the canonical K and with the settings of Section V-C, holding the data slice fixed. For each (method, seed) we compute (a) KMeans-on- θ to K clusters and report ARI against the ground-truth label, (b) NMI against the ground-truth label, and (c) c_v coherence on the inferred basis. The aggregate per-method statistic is mean $\pm$ std ARI over the five seeds, plus the same for c_v . LDA enters this comparison through its single-seed head-to-head fit on the same data slice. The seed stability of the LDA topic basis itself is measured by refitting LDA across 15 seeds and matching topic vectors between seed pairs with the Hungarian-matched cosine of F-8 (§IV-H), which takes the place of the topic-overlap score in the multi-seed stability test of Greene, O’Callaghan and Cunningham [31]; these per-scene measurements are released in `topic_stability/`.

D. F-4: capacity sensitivity

We sweep K over the grid $\{4, 6, 8, 10, 12, 16\}$ with the canonical hyperparameters and report ARI(canonical fit, KMeans on θ at K) against ground-truth label. The grid is set in data-pipeline/build_lda_sweep.py as $K_{\text{grid}} = [4, 6, 8, 10, 12, 16]$ and is mirrored by the lda_sweep/<scene>.json payloads consumed by the F-4 panel. The curve diagnoses *capacity over-/under-allocation*: capacities below n_{classes} force topic merging and lower ARI; capacities well above n_{classes} cause topic splitting and a different failure mode.

E. F-5: band-mask robustness

We refit the canonical LDA on four band-restricted vocabularies: VNIR-only ($[400, 1100]$ nm), SWIR-only ($[1100, 2500]$ nm), no-water (drop $[1350, 1430] \cup [1800, 1950] \cup [2480, 2500]$ nm), and top-50 Fisher (keep the 50 bands with highest per-band Fisher ratio against the ground-truth label). For each (scene, mask) we compute the *paired adjusted Rand index* defined in the companion conference paper, restated here for self-containment:

$$\text{ARI}_{\text{paired}} = \text{ARI}\left(\{z_d\}_{d \in \mathcal{S}}, \{z_d^{(m)}\}_{d \in \mathcal{S}}\right), \quad (6)$$

where $z_d = \arg \max_k \theta_{d,k}$ is the canonical dominant-topic map, $z_d^{(m)}$ is the masked dominant-topic map, and $\mathcal{S}$ is the set of pixels labelled in both maps.

F. F-6: cross-method agreement

Six unsupervised partitions are computed for each labelled scene: SLIC with $S = 500$ [25], SLIC with $S = 2000$, Felzenszwalb [24], 7×7 spectral patches reduced to a cluster index by KMeans, 15×15 patches by KMeans, and a trivial pixel-wise (identity) partition. We then compute the all-pairs ARI and NMI matrix between every pair of partitions including the canonical topic-dominant map and the ground-truth label, yielding an 8×8 comparison surface per scene. The diagonal is 1.0; off-diagonal entries diagnose which partitions agree with each other independently of the ground-truth label, surfacing shared inductive biases.

G. F-7: topic-label coupling

For each (scene, topic t) we compute the empirical conditional $P(L | t)$ via the confusion matrix between the dominant-topic map and the ground-truth label. The aggregate per-topic statistic is $H(L | t) = -\sum_{\ell} P(\ell | t) \log P(\ell | t)$, and a KL divergence against the marginal $P(L)$ measures how concentrated the topic's label support is. Low $H(L | t)$ topics correspond to label-aligned topics (one canonical class dominates); high $H(L | t)$ topics correspond to mixed-label regions and are useful candidates for transition zones or shadow/cloud artefacts.

H. F-8: per-topic identity tracking under masks

Given the canonical fit and a masked refit, the confusion matrix $C \in \mathbb{Z}_{\geq 0}^{K \times K^{(m)}}$ with $C_{k,j} = |\{d \in \mathcal{S} : z_d = k \wedge z_d^{(m)} = j\}|$ defines a maximum-weight bipartite matching whose solution $\sigma^* = \arg \max_{\sigma} \sum_k C_{k, \sigma(k)}$ is computed by the Hungarian algorithm of Kuhn [22] with Munkres's $O(K^3)$ refinement [32] via `scipy.optimize.linear_sum_assignment` [33]. The *swap rate under alignment* is $|\mathcal{S}|^{-1} \sum_{d \in \mathcal{S}} \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)]$, the residual disagreement that survives permutation realignment.

I. F-9: HIDSAG cross-preprocessing stability

HIDSAG mineralogical samples admit multiple preprocessing recipes (continuum-removed reflectance, derivative spectra, smoothed continuum-preserved). We refit LDA on three recipes per subset and report ARI(dominant-topic recipe A, dominant-topic recipe B) for each pair, summarising preprocessing-induced topic shifts that may dominate the underlying signal on mineralogical data.

J. F-10: cross-scene topic transfer

We project the canonical topic basis ϕ of source scene s onto the document-term matrix of target scene t , compute $\theta^{(t \leftarrow s)}$ via a single E-step, and report ARI of the resulting dominant-topic map against the ground-truth label of t . The reverse direction is also reported. The transfer ARI is informative when both scenes are sensorially similar (e.g. Salinas and Salinas-A) and useful as a negative control when they are not (e.g. Indian Pines to Pavia U).

K. F-11: rate-distortion of θ

We reconstruct each pixel spectrum from its θ coefficient and the basis ϕ as $\hat{x}_d = \phi^\top \theta_d$, scaled back from the quantised band-frequency representation. The distortion is reported as the per-pixel mean-squared error against the original spectrum, plotted against the dictionary size K (the number of basis spectra). Note this is a distortion-vs-complexity curve, not a strict rate-distortion curve: the reconstruction $\phi^\top \theta_d$ is a *soft* convex combination over all K topics, whose description cost is that of the continuous simplex vector $\theta_d \in \Delta^{K-1}$, not the $\log_2 K$ bits of a hard one-of- K codeword selection. The curve diagnoses how aggressively the topic representation compresses the spectrum while remaining usable for downstream tasks.

L. F-12: external spectral-library and method-rank alignment

For each labelled scene we align every canonical LDA topic against the USGS Spectral Library Version 7 [34] by cosine similarity, matching each per-topic reconstructed spectrum to its nearest library category and recording the alignment per topic; the external_validation/<scene>_literature.json artefacts store these per-topic matchings. On the HIDSAG mineralogical subsets, where spectral-library categories are physically meaningful, we additionally report the Friedman method-rank cross-comparison across partitions (external_validation/

`<subset>_methods.json`). This axis is a spectral-plausibility and cross-method check, not a published-accuracy benchmark. Its outputs are released in the artefact set and are not analysed further in this paper.

V. EXPERIMENTAL SETUP

A. Labelled scenes

The six standard labelled HSI benchmarks were obtained from the UPV/EHU Hyperspectral Remote Sensing Scenes collection. The scenes span three sensor families (AVIRIS, ROSIS, EO-1 Hyperion) and three application domains (agriculture, urban, wetlands), and are listed in Table II with the per-scene tokenisation outcome.

B. HIDSAG mineralogical subsets

The Hyperspectral Image Database for Supervised Analysis in Geometallurgy (HIDSAG [40], [41]) is a curated collection of hyperspectral cubes over mineralogical samples, acquired with VNIR and SWIR sensors and accompanied by mineralogical, geochemical, and geometallurgical labels. Five thematic subsets are exposed in the companion code repository: GEOMET (geometallurgical recovery + comminution targets), MINERAL1 (silicates / sulfides), MINERAL2 (high-sulfidation epithermal mineralogy), GEOCHEM (bulk geochemistry), and PORPHYRY (porphyry-copper deposits). Each subset contains 19 target labels distributed across multiple samples; the document-construction pipeline averages pixels within a region of interest to produce one spectrum per (sample, measurement, cube) tuple. The canonical fit on HIDSAG follows the same online-VB LDA recipe as for labelled scenes with the addition that HIDSAG documents are typically longer (larger $\sum_b c_{d,b}$) by construction.

C. Hyperparameters held fixed

All LDA fits in this study use the same hyperparameter slate: $K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$, $T_{\text{max}} = 60$ online-VB passes, learning offset 10.0, learning decay 0.7, batch size 128, seed 42, samples per class 220. ProdLDA and ETM are the two amortised models stated in Section II: the mixture-decoder form labelled ProdLDA and ETM, sharing the $\mathcal{N}(0, I)$ prior and an encoder with two hidden layers of 100 units, with ETM word embeddings of dimension 64. Both are trained with Adam at learning rate 10^{-3} for 80 epochs with batch size 128, K matched to the LDA canonical, and seeds $\{0, 1, 2, 3, 4\}$ for the seed-stability sweep. Their documents are band-frequency counts at scale 20 (each min–max-normalised spectrum of the 220-per-class sample multiplied by 20 and rounded) rather than the 13-level canonical tokenisation. The hierarchical Bayesian classifier of F-1 uses NUTS with 1000 draws, 1000 tune samples, 2 chains and a target acceptance of 0.9; the archived fits record no divergence or convergence diagnostics.

VI. RESULTS

A. F-1: classification across method families

Figure 1 and Table III report the five methods on the labelled-scene block. The topic-routed soft classifier exceeds the raw-spectrum logistic baseline by 0.005 macro-F1 (0.915 against 0.910; 94% paired bootstrap interval 0.003 to 0.007, higher on 20 of the 30 cells), a small difference that the hierarchical model, which does not pair the cells, places at $P[\mu_{\text{soft}} > \mu_{\text{raw}}] = 0.641$. The theta-only logistic regression trails the baseline by 0.325 on every one of the 30 cells, confirming that θ alone discards information that the raw spectrum still contains; the routed classifiers recover it by keeping the raw spectrum as the input of each topic’s expert.

The HIDSAG block (Table IV) repeats the θ -only result. The topic-logistic classifier trails the raw-logistic baseline by 0.268 macro-F1 (0.430 against 0.699; lower on 18 of the 19 targets; $P[\mu_{\text{topic}} > \mu_{\text{raw}}] = 0.0$), and the cube-topic and region-topic variants (both using a coarser, region-pooled θ) are no better than the per-spectrum topic posterior. On mineralogical SWIR data, as on the labelled scenes, the topic proportions *discard* information that a multinomial logistic on the raw spectrum exploits. This study has no HIDSAG results for the topic-routed classifiers, so it cannot say whether routing the raw spectrum through the topics would close the gap there.

B. F-2 + F-3: coherence and seed stability of LDA, ProdLDA, ETM

Table V reports the seed-stability statistics on Indian Pines for the three methods at $K = 12$, $n_{\text{seeds}} = 5$.

The seed-stability picture is more nuanced than the head-to-head ARI ranking suggests. ProdLDA has the highest coherence ($c_v = 0.630$) and the lowest seed-to-seed variance ($\sigma_{\text{ARI}} = 0.003$) but the lowest cluster ARI against label. Among the neural variants, ETM reaches the highest cluster ARI (0.257) at a much lower coherence. LDA without amortisation reaches the highest single-seed ARI (0.287 on the head-to-head fit) but is not directly comparable to the multi-seed mean- $\pm$ -std statistics of the neural variants. The methodological consequence: *ranking by ARI alone inverts the ranking by coherence*, and a paper that picks one of the two metrics in isolation will reach a different methodological conclusion.

C. F-5: band-mask robustness

Table VI and Fig. 2 report the paired ARI across (scene, mask) tuples on the six labelled scenes. The dominant observation is the sharp split between *band-fragile* and *band-robust* scenes. KSC paired ARI ranges from 0.007 to 0.013 across all four masks: the masked LDA assigns pixels to topics with effectively chance-level agreement with the canonical fit. Botswana sits in the same regime only under SWIR (0.013); under VNIR (0.108), no-water (0.127) and top-50 Fisher (0.117) it reaches 0.11–0.13, still well below the labelled-scene average. The canonical-fit topic identity on these scenes is *not* recoverable from any reasonable band subset. Salinas-A under SWIR-only restriction reaches paired ARI = 0.766:

TABLE II

LABELLED HYPERSPECTRAL SCENES USED IN THIS STUDY. B DENOTES THE BAND COUNT AFTER CORRECTION; D IS THE NUMBER OF DOCUMENTS (SAMPLED PIXELS) ENTERING THE CANONICAL FIT; K IS THE CANONICAL TOPIC COUNT. INDIAN PINES, SALINAS, AND SALINAS-A USE THE AVIRIS SENSOR; PAVIA U IS ROSIS-03; KSC IS AVIRIS-OLD; BOTSWANA IS EO-1 HYPERION.

Scene	Sensor	Spatial	B	D	K	Reference
Indian Pines	AVIRIS	145×145	200	2812	12	[35]
Salinas	AVIRIS	512×217	204	3520	12	[36]
Salinas-A	AVIRIS	83×86	204	1320	6	[36]
Pavia University	ROSIS-03	610×340	103	1980	9	[37]
Kennedy Space C.	AVIRIS	512×614	176	2686	12	[38]
Botswana	EO-1 H.	1476×256	145	2775	12	[39]

TABLE III

F-1 ON THE LABELLED-SCENE BLOCK (SIX SCENES $\times$ FIVE FOLDS $\times$ FIVE METHODS = 150 OBSERVATIONS): OBSERVED MEAN MACRO-F1, MEAN PAIRED DIFFERENCE Δ FROM RAW_LOGISTIC WITH ITS 94% BOOTSTRAP INTERVAL OVER THE 30 CELLS, AND $P = P[\mu_m > \mu_{\text{RAW}}]$ FROM THE HIERARCHICAL MODEL. THE MODEL'S POSTERIOR LOCATIONS μ_m ARE NOT LISTED: THEY ARE IDENTIFIED ONLY UP TO THE SCENE OFFSETS (§IV-A). ALL PAIRWISE PROBABILITIES ARE IN THE COMPANION ARTEFACT (CROSS_CLASSIFICATION_BAYESIAN.JSON).

Method	mean	Δ	94% interval	P
topic_routed_soft	0.915	+0.005	[+0.003, +0.007]	0.641
raw_logistic	0.910	ref.	—	—
topic_routed_hard	0.909	-0.001	[-0.008, +0.006]	0.464
pca_K_logistic	0.847	-0.064	[-0.099, -0.031]	0.000
theta_logistic	0.586	-0.325	[-0.359, -0.289]	0.000

TABLE IV

F-1 ON THE HIDSAG BLOCK (19 TARGETS ACROSS FIVE SUBSETS $\times$ FIVE METHODS = 95 OBSERVATIONS), IN THE FORMAT OF TABLE III; INTERVALS OVER THE 19 PAIRED TARGETS.

Method	mean	Δ	94% interval	P
raw_logistic	0.699	ref.	—	—
pca_logistic	0.674	-0.025	[-0.050, -0.003]	0.284
topic_logistic	0.430	-0.268	[-0.374, -0.176]	0.000
region_topic_logistic	0.392	-0.307	[-0.409, -0.212]	0.000
cube_topic_logistic	0.360	-0.339	[-0.437, -0.246]	0.000

the masked fit recovers nearly the same partition (Fig. 3), indicating that the spectral signature carried by the SWIR bands is sufficient to identify the canonical structure of this scene.

D. F-6: cross-method clustering agreement

Table VII reports the all-pairs ARI matrix on Indian Pines. The dominant observation: the canonical topic-dominant map has *low* pairwise ARI against the spatial-segmentation partitions (SLIC, Felzenszwalb, patches) but the label–topic ARI is 0.203, comparable to the SLIC-500-vs-label ARI 0.227. The two partitions are arriving at similar accuracy levels through different inductive biases; the cross-method agreement matrix surfaces this in a way that the per-method OA bar chart hides.

E. F-7 + F-8: topic–label coupling and per-topic identity

We use the per-topic KL divergence $\text{KL}(P(L | t) \| P(L))$ between the empirical distribution of ground-truth labels conditioned on topic t and the marginal label distribution as a per-topic concentration measure; the within-scene mean and

TABLE V

F-3 SEED STABILITY ON INDIAN PINES, $K = 12$, $n_{\text{SEEDS}} = 5$. KMEANS-ON- θ VS GROUND-TRUTH LABEL.

Method	ARI mean	ARI std	c_v mean	c_v std
LDA	0.287	—	0.450	—
ProdLDA	0.165	0.003	0.630	0.012
ETM	0.257	0.003	0.357	0.019

max over the canonical-fit topics are reported in Table VIII, computed from the `kl_to_label_prior_per_topic` field of `topic_to_data/<scene>.json`.

F. Combining F-5 and F-1: band-fragile or label-recoverable?

A scene's appearance in the band-fragile regime (paired ARI ≈ 0) does not, by itself, imply low classifier accuracy on that scene. Botswana paired ARI is ≈ 0.01 on SWIR but the mask-vs-label ARI on the same fit is 0.098 (within range of labelled-scene comparable values). The two metrics are measuring different things: paired ARI(canonical, masked) asks whether the canonical topic identities survive a band restriction, while ARI(masked, label) asks whether the masked fit still discriminates the classes. A band-fragile scene with a non-zero mask-vs-label ARI is one on which the topic basis is being *re-derived from scratch* under each mask, with topic identity unstable but topic discriminability preserved.

G. Synthesis: per-scene model card

Table IX consolidates the per-scene values across axes F-1, F-2, F-3, F-5, F-7 for the labelled scenes. We argue that this synthesis, not the single OA number, is the appropriate unit of reporting for a topic model on HSI.

The F-1 column spans 0.819 (Pavia University) to 0.996 (Salinas-A), and it does not track band robustness: the two band-fragile scenes, Kennedy Space Center and Botswana (F-5 of 0.013 and 0.127), have the fourth- and second-highest F-1.

VII. DISCUSSION

A. Why the band-mask diagnostic separates scenes

The split between *band-fragile* (KSC, Botswana) and *band-robust* (Salinas-A) scenes is, on its face, surprising: KSC and Botswana have rich spectral signatures, broad spatial coverage,

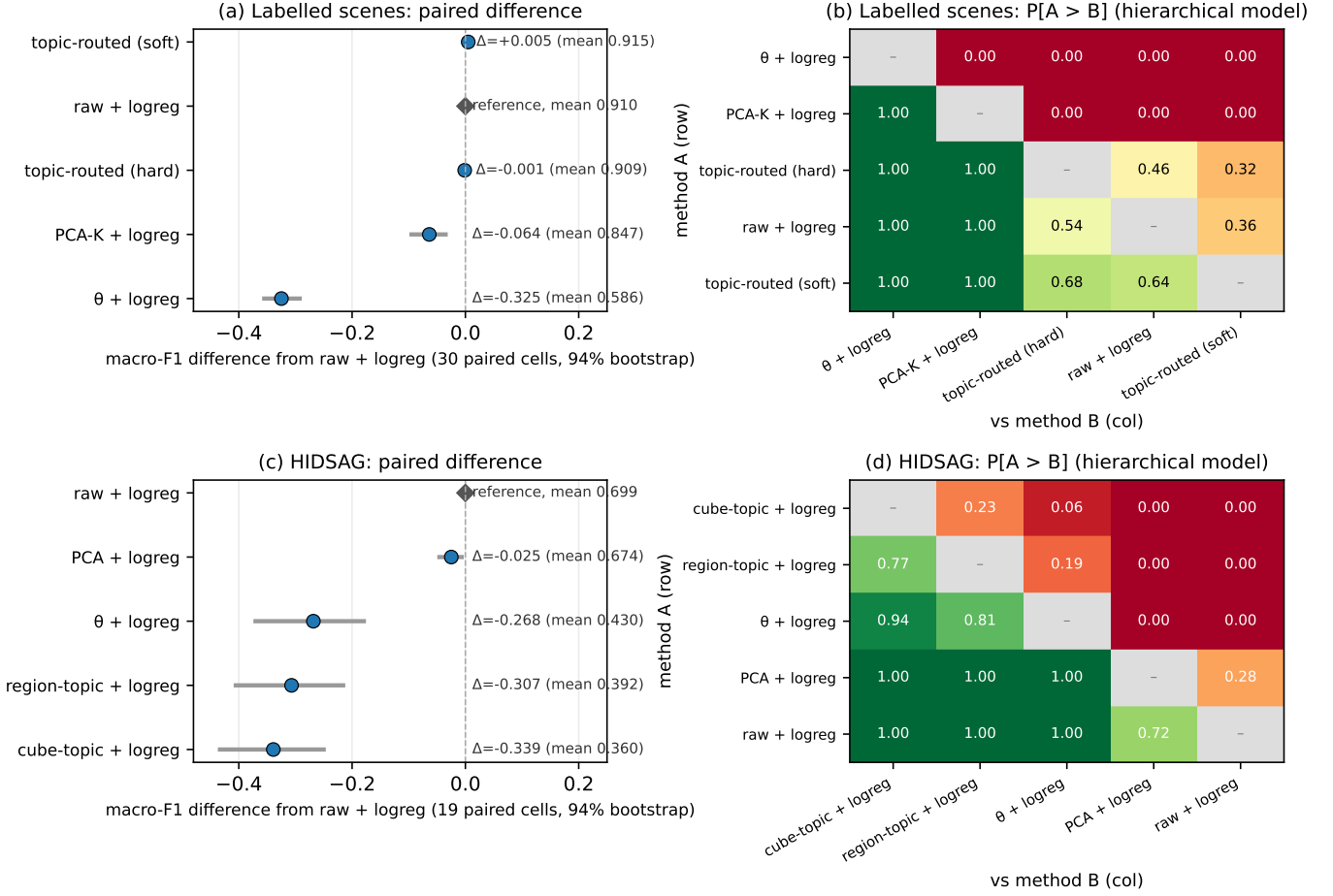
Method comparison on axis F-1: paired macro-F1 differences and hierarchical-model $P[A > B]$ 

Fig. 1. F-1 method comparison. (a) Labelled scenes (6 scenes $\times$ 5 folds): mean paired macro-F1 difference of each method from `raw_logistic` over the 30 (scene, fold) cells, with its 94% bootstrap interval; the text gives the difference and the observed mean. (b) Pairwise $P[A > B]$ from the hierarchical Bayesian model (NUTS, 1000 draws, 1000 tune, 2 chains). (c), (d) HIDSAG mineralogical subsets (19 targets across 5 subsets): the same two views. On both blocks the logistic regressions on topic proportions (θ , cube-topic, region-topic) sit 0.27 to 0.34 below `raw_logistic`; on the labelled scenes `topic_routed_soft`, which keeps the raw spectrum, sits 0.005 above it.

and a comparable per-class sample budget to Salinas-A. Two mechanisms are consistent with the diagnostic.

Mechanism (i): canonical topics exploit band-level redundancy. The canonical fit may be using fine-grained spectral co-variation patterns that are present in the full-band representation and disappear when half the bands are dropped. Under this mechanism the canonical basis is not *wrong*, but it is *not transportable* across spectral windows. Any claim of the form “topic k corresponds to a specific land-cover signature” presupposes a fixed reference frame in which ϕ_k is defined; the diagnostic shows that on KSC and Botswana this reference frame is band-window-conditional.

Mechanism (ii): the canonical topics are weakly identified. KSC has 13 classes and, at the canonical 220 samples per class, $D = 2686$ documents; Botswana has 14 classes and $D = 2775$. The latent topic basis ϕ has $K \times B \approx 12 \times 176$ free parameters on KSC. With $D = 2686$ and a per-document token budget bounded by 13 quantisation levels times B bands, the data-to-parameter ratio is unfavourable; the posterior over

ϕ has substantial mass over many high-likelihood configurations, and a band restriction selects a different posterior mode of the same family.

The diagnostic does not distinguish (i) from (ii); both are valid warnings to a downstream consumer of the topic basis. The framework’s contribution is to make the warning legible.

B. What the topic proportions lose

A logistic regression on the topic proportions trails the raw-spectrum baseline on both benchmark families: by 0.325 macro-F1 on the labelled scenes (every one of the 30 scene-fold cells) and by 0.268 on the HIDSAG targets (18 of 19). The deficit is therefore not specific to mineralogical data. A θ of at most twelve proportions summarises a spectrum of B bands, and the narrow, locally discriminative absorption features that separate classes (mineral classes on HIDSAG) plausibly do not survive that compression; the raw multinomial logistic keeps them. On the labelled scenes the topic-routed classifiers, which use θ only to route the raw spectrum to per-topic experts,

TABLE VI

F-5 PAIRED ARI PER (SCENE, MASK) TUPLE. BOLD CELLS INDICATE THE BAND-MASK THAT ACHIEVES THE HIGHEST PAIRED ARI PER SCENE. PAVIA U $\times$ SWIR IS AUTO-SKIPPED (SENSOR IS VNIR-ONLY). HUNGARIAN-ALIGNED SWAP RATES AND PER-TOPIC ALIGNMENTS ARE IN THE COMPANION ARTEFACT `BAND_MASKS/CANONICAL_COMPARISON.JSON`.

Scene	Paired ARI				Bands kept of B			
	vnir	swir	no-water	top-50	vnir	swir	no-water	top-50
Indian Pines	0.427	0.527	0.499	0.398	67/200	133/200	177/200	50/200
Salinas	0.540	0.435	0.502	0.507	68/204	136/204	180/204	50/204
Salinas-A	0.521	0.766	0.533	0.352	68/204	136/204	180/204	50/204
Pavia University	0.415	—	0.415	0.387	103/103	0/103	103/103	50/103
Kennedy Space C.	0.009	0.011	0.007	0.013	59/176	117/176	155/176	50/176
Botswana	0.108	0.013	0.127	0.117	49/145	97/145	127/145	50/145

TABLE VII

F-6 CROSS-METHOD ARI MATRIX ON INDIAN PINES. DIAGONAL IS 1.0 (OMITTED). EACH ENTRY IS ARI(ROW, COLUMN). LABEL DENOTES THE GROUND-TRUTH CLASS MAP; TOPIC-DOMINANT DENOTES THE CANONICAL LDA DOMINANT-TOPIC MAP. SLIC SIZES CORRESPOND TO TARGET SEGMENT COUNTS; PATCHES DENOTE KMEANS ON FLATTENED 7×7 OR 15×15 SPECTRAL WINDOWS; PIXEL IS THE TRIVIAL IDENTITY PARTITION.

	label	topic-dom.	felz.	SLIC-500	SLIC-2000	patch-15	patch-7
label	—	0.203	0.567	0.227	0.051	0.281	0.117
topic-dominant	0.203	—	0.120	0.039	0.010	0.047	0.021
felzenszwalb	0.567	0.120	—	0.306	0.076	0.372	0.170
SLIC-500	0.227	0.039	0.306	—	0.239	0.359	0.374
SLIC-2000	0.051	0.010	0.076	0.239	—	0.150	0.316
patch-15	0.281	0.047	0.372	0.359	0.150	—	0.292
patch-7	0.117	0.021	0.170	0.374	0.316	0.292	—

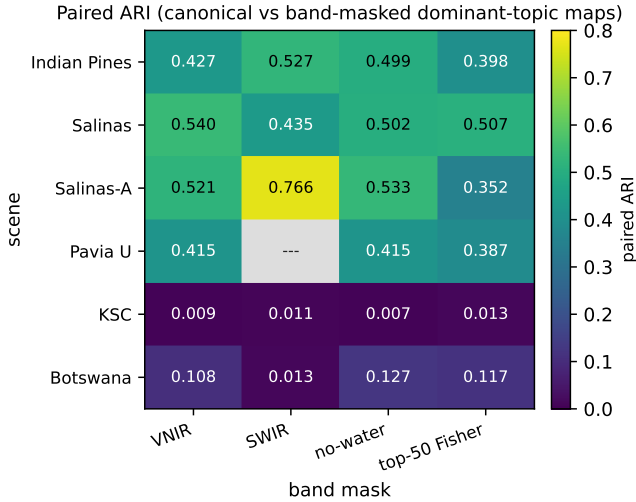


Fig. 2. F-5 paired ARI heatmap across (scene, mask) tuples. KSC paired ARI sits at 0.007–0.013 across all four masks (band-fragile regime); Botswana shows a mixed pattern (SWIR 0.013 band-fragile; VNIR 0.108, no-water 0.127, top-50 Fisher 0.117 indicating partial structure under those masks); Salinas-A under SWIR-only reaches 0.766 (band-robust regime). The remaining tuples fall in the partial-robustness range 0.35–0.54. Pavia U $\times$ SWIR is auto-skipped (sensor is VNIR-only).

match the baseline (a difference of +0.005 for soft gating and −0.001 for hard gating). The implication is that the topic representation is an interpretable routing and summary layer rather than a replacement for the spectrum as classifier input. Whether routing closes the gap on HIDSAG as it does on the labelled scenes was not tested and is a direct follow-up.

TABLE VIII

F-7 TOPIC-LABEL COUPLING: PER-TOPIC $KL(P(L | t) || P(L))$ MEAN AND MAX ON THE *canonical fit* OF EACH LABELLED SCENE (I.E. THE NO-MASK LDA FIT). HIGHER VALUES DENOTE MORE SHARPLY LABEL-ALIGNED TOPICS.

Scene	KL mean (nats)	KL max (nats)
Indian Pines	1.35	2.51
Salinas	1.73	2.77
Salinas-A	1.29	1.79
Pavia University	0.81	1.17
Kennedy Space C.	0.95	2.50
Botswana	1.73	3.37

C. LDA versus ProdLDA versus ETM

The ARI-vs- c_v trade-off in Section VI-B bears on the criterion on which ProdLDA was introduced as an improvement over LDA, higher topic coherence on text corpora [10], a claim made for its product-of-experts decoder. On Indian Pines the amortised mixture-decoder model labelled ProdLDA here also has the highest coherence (ProdLDA $c_v = 0.630$ vs LDA $c_v = 0.450$ vs ETM $c_v = 0.357$), but the cluster-ARI ordering inverts (LDA 0.287 vs ETM 0.257 vs ProdLDA 0.165). The coherence-vs-clustering disagreement is, on HSI, large enough to flip the practitioner-level recommendation: a practitioner who picks ProdLDA because its coherence is highest will obtain a topic basis that clusters worst against ground truth. We therefore recommend, for HSI specifically, reporting both metrics side-by-side rather than choosing one as representative.

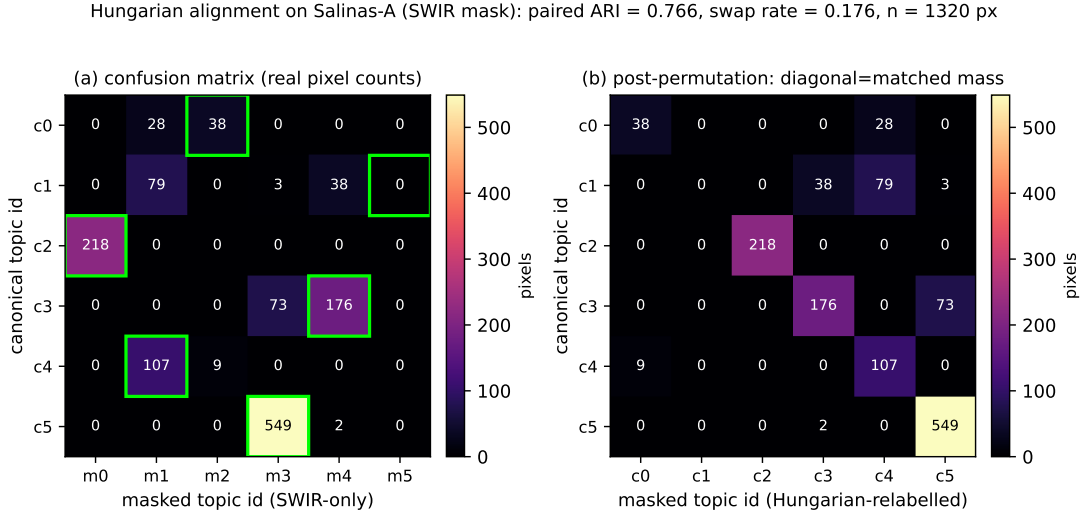


Fig. 3. F-8 Hungarian topic-id alignment on Salinas-A under SWIR-only restriction. Lime rectangles mark the maximum-weight bipartite match σ^* ; the diagonal under permutation dominates, consistent with the high paired ARI (0.766) and the low swap rate (0.176) on this (scene, mask) tuple.

TABLE IX

PER-SCENE MODEL CARD SYNTHESISING THE FRAMEWORK’S MAIN AXES. F-1 COLUMN: MACRO-F1 OF TOPIC_ROUTED_SOFT (THE CANONICAL TOPIC-BASED CLASSIFIER), MEAN OF THE FIVE FOLDS. F-2: c_v COHERENCE OF THE CANONICAL FIT. F-3: SINGLE-SEED HEAD-TO-HEAD KMEANS-VS-LABEL ARI OF LDA. F-5: MAX PAIRED ARI ACROSS THE FOUR MASKS (THE “BEST-CASE BAND-ROBUSTNESS”). F-7: PER-TOPIC $KL(P(L | t) || P(L))$ MEAN OVER CANONICAL-FIT TOPICS.

Scene	F-1 (macro-F1)	F-2 (c_v)	F-3 (ARI, 1 seed)	F-5 (max paired)	F-7 (KL nats)
Indian Pines	0.839	0.450	0.287	0.527	1.35
Salinas	0.954	—	—	0.540	1.73
Salinas-A	0.996	—	—	0.766	1.29
Pavia University	0.819	—	—	0.415	0.81
Kennedy Space C.	0.921	—	—	0.013	0.95
Botswana	0.962	—	—	0.127	1.73

D. Limitations

The framework has three operative limitations. *First*, the canonical fit uses a single K per scene, chosen by the formula $K = \max(4, \min(12, n_{\text{classes}}))$. A more sensitive analysis would sweep K and report per-axis behaviour against K ; we report the K -sweep on F-4 but other axes are computed at the canonical K . *Second*, the tokenisation $Q + 1 = 13$ levels is uniform across scenes, not adapted to per-scene spectral dynamic range. *Third*, F-12 matches topics to library categories by cosine similarity alone: a high cosine indicates spectral plausibility, not a mineralogical identification, and the paper does not compare against published accuracies, whose train/test splits would not match the splits used here.

VIII. REPRODUCIBILITY ARTEFACTS

The full reproducibility package consists of:

- **Builder source.** Forty-plus deterministic Python builders under `data-pipeline/builders/`, each accepting a fixed seed and writing a deterministic JSON or NPZ artefact under `data/derived/`. Each builder records its `builder_version` and `generated_at` ISO timestamp in the artefact metadata.
- **Derived artefacts.** 3895 JSON and binary files, 449 MB total, including: per-scene canonical fits

(`theta_grid.bin`, `dominant_topic_map.bin`), the band-mask canonical comparison (`band_masks/canonical_comparison.json`, the source of all F-5 numbers above), the cross-method agreement matrices (`cross_method_agreement/`), the seed-stability sweeps (`neural_topic_seed_stability/`), the hierarchical Bayesian posteriors (`method_statistics_labelled/cross_classification_bayesian.json` and `method_statistics_hidsag/cross_classification_bayesian.json`), and the HIDSAG cross-preprocessing stability sweeps.

- **Web application.** A FastAPI backend serving the artefacts over 59 REST endpoints, plus a React+Vite frontend at <https://lda-hsi.fasl-work.com> that consumes them as interactive visualisations.
- **Endpoint test harness.** A test script covering 109 endpoints that verifies endpoint health, response schemas, and the content of the single-page-application shell at each deployment.
- **Companion paper.** The F-5 band-mask diagnostic is reported in detail in [42].

All code is MIT-licensed at <https://github.com/>

fsantibanezleal/CAOS_LDA_HSI.

IX. CONCLUSION

The accuracy-only view of topic models on hyperspectral imagery hides the questions an interpretability-motivated practitioner most needs answered: is the topic basis reproducible across seeds, robust to spectral-window choice, calibrated against label entropy, actually outperforming linear baselines, and consistent across preprocessing recipes? We propose a twelve-axis evaluation framework that returns one operational metric per axis and stores each metric as a deterministic JSON artefact. Applied to six labelled HSI benchmarks and five HIDSAG mineralogical subsets, the framework surfaces three findings the accuracy-only view does not: the topic-routed soft classifier matches the raw-spectrum logistic baseline on the labelled scenes (0.915 against 0.910 macro-F1); the band-mask diagnostic separates scenes whose canonical topics are band-robust (Salinas-A SWIR paired ARI 0.766) from scenes whose topics are band-fragile (KSC under every mask and Botswana under SWIR, paired ARI $\approx$ 0.01); and a classifier on the topic proportions alone trails the raw-spectrum baseline on both the labelled scenes (by 0.325 macro-F1) and the HIDSAG targets (by 0.268). The full 3895-artefact reproducibility package is released under MIT licence.

Future work includes (i) a per-axis K -sweep extending the single- K canonical fit, (ii) per-scene tokenisation budgets adapted to spectral dynamic range, (iii) extending the framework to neural unmixing variants (autoencoder unmixing, transformer-based topic models), and (iv) developing an information-theoretic within-mask Fisher complement to F-5 that does not depend on the unmasked-spectrum ranking.

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CONFLICT OF INTEREST

The author is a co-author of the HIDSAG dataset descriptor publication [40] and the accompanying data release [41], and first author of the earlier conference paper introducing the LDA-V1/V2/V3 corpus recipes and topic-routed regressor on private geometallurgical data [46]. The present paper is methodologically distinct: it proposes a twelve-axis evaluation framework for topic models on hyperspectral imagery and applies the framework, among other benchmarks, to the HIDSAG mineralogical subsets. HIDSAG is cited as a data source, not as a methodological contribution of this work. The author declares no other conflict of interest.

DATA AVAILABILITY STATEMENT

All numerical results in this paper derive from deterministic artefacts under `data/derived/` in the companion code repository at https://github.com/fsantibanezleal/CAOS_LDA_HSI. The paired F-1 comparisons of Tables III and IV are computed from `topic_routed_classifier/` and `core/local_core_benchmarks.json` by `figures/source/build_bayesian_method_comparison.py` of the manuscript repository https://github.com/fsantibanezleal/CAOS_LDA_HSI_Paper. The HIDSAG database is available from [41]; the labelled hyperspectral benchmark scenes (Indian Pines, Salinas, Salinas-A, Pavia U, KSC, Botswana) are available from the UPV/EHU Hyperspectral Remote Sensing Scenes collection [35]–[39]. The 3895-artefact derived bundle, the `build_*.py` pipeline source, the FastAPI backend, and the React frontend are released under the MIT licence. The interactive demonstration is hosted at <https://lda-hsi.fasl-work.com>.

REFERENCES

- [1] A. Plaza, J. A. Benediktsson, J. W. Boardman, J. Brazile, L. Bruzzone, G. Camps-Valls, J. Chanussot, M. Fauvel, P. Gamba, A. Gualtieri, M. Marconcini, J. C. Tilton, and G. Trianni, "Recent advances in techniques for hyperspectral image processing," *Remote Sensing of Environment*, vol. 113, no. Supplement 1, pp. S110–S122, 2009.
- [2] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Processing Magazine*, vol. 31, no. 1, pp. 45–54, 2014.
- [3] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 43, no. 4, pp. 898–910, 2005.
- [4] J. M. Bioucas-Dias, A. Plaza, N. Dobigeon, M. Parente, Q. Du, P. Gader, and J. Chanussot, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 5, no. 2, pp. 354–379, 2012.
- [5] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 7, no. 6, pp. 2094–2107, 2014.
- [6] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 158, pp. 279–317, 2019.
- [7] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent Dirichlet allocation," *Journal of Machine Learning Research*, vol. 3, pp. 993–1022, 2003. [Online]. Available: <https://jmlr.org/papers/v3/blei03a.html>
- [8] M. D. Hoffman, D. M. Blei, and F. R. Bach, "Online learning for latent Dirichlet allocation," in *Advances in Neural Information Processing Systems 23 (NIPS 2010)*, 2010, pp. 856–864. [Online]. Available: <https://papers.nips.cc/paper/3902-online-learning-for-latent-dirichlet-allocation>
- [9] S. Zou and A. Zare, "Hyperspectral unmixing with endmember variability using partial membership latent Dirichlet allocation," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2017, pp. 6200–6204.
- [10] A. Srivastava and C. Sutton, "Autoencoding variational inference for topic models," in *International Conference on Learning Representations (ICLR)*, 2017, arXiv:1703.01488. [Online]. Available: <https://openreview.net/forum?id=BybtVK9lg>
- [11] A. B. Dieng, F. J. R. Ruiz, and D. M. Blei, "Topic modeling in embedding spaces," *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 439–453, 2020.
- [12] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *International Conference on Learning Representations (ICLR)*, 2014. [Online]. Available: <https://arxiv.org/abs/1312.6114>

- [13] Y. Miao, L. Yu, and P. Blunsom, "Neural variational inference for text processing," in *International Conference on Machine Learning (ICML)*, 2016, pp. 1727–1736. [Online]. Available: <http://proceedings.mlr.press/v48/miao16.html>
- [14] M. Wahabzada, A.-K. Mahlein, C. Bauckhage, U. Steiner, E.-C. Oerke, and K. Kersting, "Plant phenotyping using probabilistic topic models: Uncovering the hyperspectral language of plants," *Scientific Reports*, vol. 6, p. 22482, 2016.
- [15] R. A. Borsoi, T. Imbiriba, J. C. M. Bermudez, C. Richard, J. Chanussot, L. Drumetz, J.-Y. Tourneret, A. Zare, and C. Jutten, "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geoscience and Remote Sensing Magazine*, vol. 9, no. 4, pp. 223–270, 2021.
- [16] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, and A. Lerchner, "beta-VAE: Learning basic visual concepts with a constrained variational framework," in *International Conference on Learning Representations (ICLR)*, 2017. [Online]. Available: <https://openreview.net/forum?id=Sy2fzU9gl>
- [17] D. Card, C. Tan, and N. A. Smith, "Neural models for documents with metadata," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018, pp. 2031–2040.
- [18] D. Mimno and A. McCallum, "Topic models conditioned on arbitrary features with Dirichlet-multinomial regression," in *Proceedings of the 24th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2008, pp. 411–418, arXiv:1206.3278. [Online]. Available: <https://arxiv.org/abs/1206.3278>
- [19] M. Röder, A. Both, and A. Hinneburg, "Exploring the space of topic coherence measures," in *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining (WSDM)*, 2015, pp. 399–408.
- [20] L. Hubert and P. Arabie, "Comparing partitions," *Journal of Classification*, vol. 2, no. 1, pp. 193–218, 1985.
- [21] N. X. Vinh, J. Epps, and J. Bailey, "Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance," *Journal of Machine Learning Research*, vol. 11, pp. 2837–2854, 2010. [Online]. Available: <https://jmlr.org/papers/v11/vinh10a.html>
- [22] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.
- [23] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d'Alché Buc, E. Fox, and H. Larochelle, "Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program)," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021. [Online]. Available: <https://jmlr.org/papers/v22/20-303.html>
- [24] P. F. Felzenszwalb and D. P. Huttenlocher, "Efficient graph-based image segmentation," *International Journal of Computer Vision*, vol. 59, no. 2, pp. 167–181, 2004.
- [25] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 34, no. 11, pp. 2274–2282, 2012.
- [26] A. Gersho and R. M. Gray, *Vector Quantization and Signal Compression*, ser. The Springer International Series in Engineering and Computer Science. Boston, MA: Kluwer Academic Publishers, 1992, vol. 159.
- [27] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: <https://jmlr.org/papers/v12/pedregosa11a.html>
- [28] A. Vehtari, A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner, "Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC (with discussion)," *Bayesian Analysis*, vol. 16, no. 2, pp. 667–718, 2021. [Online]. Available: <https://doi.org/10.1214/20-BA1221>
- [29] D. Newman, J. H. Lau, K. Grieser, and T. Baldwin, "Automatic evaluation of topic coherence," in *Human Language Technologies: The 2010 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, 2010, pp. 100–108. [Online]. Available: <https://aclanthology.org/N10-1012/>
- [30] D. Mimno, H. M. Wallach, E. Talley, M. Leenders, and A. McCallum, "Optimizing semantic coherence in topic models," in *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2011, pp. 262–272. [Online]. Available: <https://aclanthology.org/D11-1024/>
- [31] D. Greene, D. O'Callaghan, and P. Cunningham, "How many topics? Stability analysis for topic models," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML-PKDD)*, ser. Lecture Notes in Computer Science, vol. 8724. Springer, 2014, pp. 498–513. [Online]. Available: <https://arxiv.org/abs/1404.4606>
- [32] J. Munkres, "Algorithms for the assignment and transportation problems," *Journal of the Society for Industrial and Applied Mathematics*, vol. 5, no. 1, pp. 32–38, 1957.
- [33] SciPy Developers, "scipy.optimize.linear_sum_assignment: SciPy reference," Online; SciPy v1.13 reference manual, 2024. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linear_sum_assignment.html
- [34] R. F. Kokaly, R. N. Clark, G. A. Swayze, K. E. Livo, T. M. Hoefen, N. C. Pearson, R. A. Wise, W. M. Benzel, H. A. Lowers, R. L. Driscoll, and A. J. Klein, "USGS spectral library version 7," U.S. Geological Survey, Data Series 1035, 2017. [Online]. Available: <https://doi.org/10.3133/ds1035>
- [35] M. F. Baumgardner, L. L. Biehl, and D. A. Landgrebe, "220 band AVIRIS hyperspectral image data set: June 12, 1992 Indian Pine test site 3," Purdue University Research Repository, 2015, Indian Pines AVIRIS hyperspectral dataset, calibrated 220-band reflectance, ground-truth labels for 16 land-cover classes.
- [36] NASA/JPL Airborne Visible/Infrared Imaging Spectrometer Project, "AVIRIS hyperspectral imagery: Salinas Valley, California," 1998, 224-band hyperspectral cube, 3.7 m spatial resolution. Calibrated and corrected version maintained by the Computational Intelligence Group (GIC) of the University of the Basque Country (UPV/EHU). Salinas-A is a 83×86 sub-scene with 6 classes. [Online]. Available: https://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Salinas_scene
- [37] German Aerospace Center (DLR) ROSIS sensor, "ROSIIS Pavia University hyperspectral dataset," 2003, Reflective Optics System Imaging Spectrometer airborne acquisition over Pavia, Italy. 103 bands after bad-band removal, 9 classes, 610×340 pixels at 1.3 m resolution. [Online]. Available: https://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_University_scene
- [38] NASA/JPL Airborne Visible/Infrared Imaging Spectrometer Project, "AVIRIS Kennedy Space Center hyperspectral imagery," 1996, Kennedy Space Center, Florida, 18 m resolution, 224 bands, 13 land-cover classes after labelling. [Online]. Available: https://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Kennedy_Space_Center_28KSC.29
- [39] NASA EO-1 Hyperion Mission, "EO-1 Hyperion Okavango Botswana hyperspectral imagery," 2001, Hyperion satellite-borne hyperspectral sensor, Okavango Delta, Botswana. 145 bands after preprocessing, 14 land-cover classes. [Online]. Available: https://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Botswana
- [40] A. Ehrenfeld, Á. F. Egaña, F. Santibáñez-Leal, F. Garrido, M. Ojeda, B. Townley, and F. Navarro, "HIDSAG: Hyperspectral image database for supervised analysis in geometalurgy," *Scientific Data*, vol. 10, no. 1, p. 164, 2023. [Online]. Available: <https://doi.org/10.1038/s41597-023-02061-x>
- [41] A. Ehrenfeld, "HIDSAG: Hyperspectral image database for supervised analysis in geometalurgy," figshare Collection, <https://doi.org/10.6084/m9.figshare.c.5983921.v1>, 2023, data collection accompanying [40]; five thematic subsets (GEOMET, MINERAL1, MINERAL2, GEOCHEM, PORPHYRY). Reference implementation at <https://github.com/alges/hidsag>.
- [42] F. Santibáñez-Leal, "A band-mask robustness diagnostic for latent Dirichlet allocation on hyperspectral imagery," Preprint, Zenodo, 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.21504109>
- [43] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. F. del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, and T. E. Oliphant, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.
- [44] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, Í. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, and P. van Mulbregt, "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, no. 3, pp. 261–272, 2020.

- [45] R. Řehůřek and P. Sojka, “Software framework for topic modelling with large corpora,” in *Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks*, 2010, pp. 45–50. [Online]. Available: <http://is.muni.cz/publication/884893/en>
- [46] F. A. Santibañez-Leal, Á. F. Egaña, A. Ehrenfeld, F. Navarro, and B. Townley, “Geometallurgical estimation of mineral samples from hyperspectral images and statistical topic modelling,” in *Proceedings of the 18th International Conference on Mineral Processing and Geometallurgy (Procemin Geomet 2022)*. Santiago, Chile: Gecamin, 2022. [Online]. Available: <https://www.researchgate.net/publication/369708272>